

$$x_3 \quad x_d$$

$$x_3 \quad \frac{-s}{d} \quad F \quad F$$

$$F$$

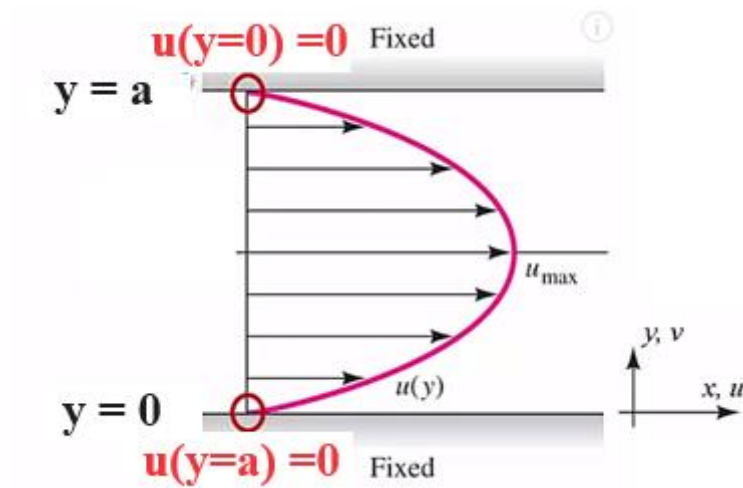
$$x_d \quad \frac{-s}{d} \quad F \quad d$$

$$F \quad \frac{-s \, d}{d}$$

$$F \quad F$$

$$x \quad \frac{-s}{d} \quad \frac{-s \, d}{d}$$

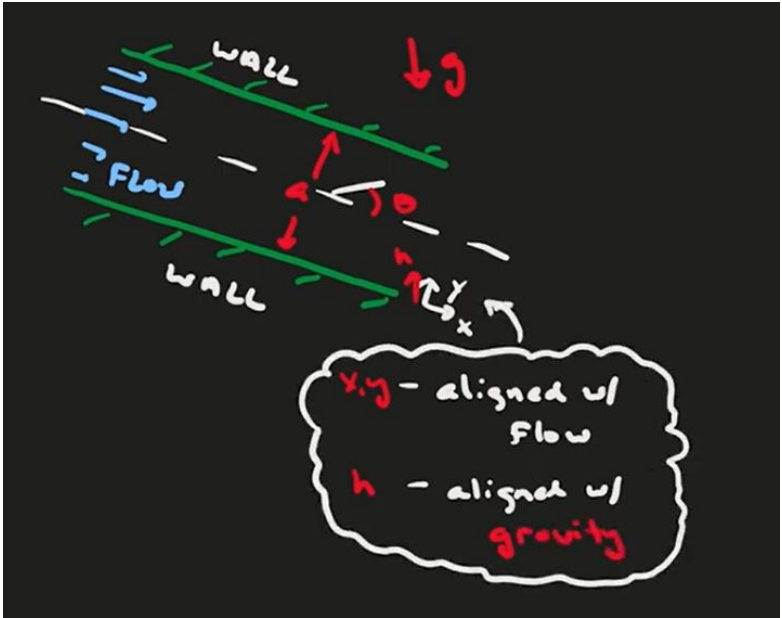
$$x \quad \frac{-s}{d} \quad d$$



Escoamento plenamente desenvolvido dominado pela força de corpo

d

d



τ νw

$$\frac{\tau}{w}$$

$$\frac{x}{z}$$

$$\frac{\tau}{z}$$

$$\frac{Q}{z} \quad \frac{z}{d}$$

$$\frac{x}{z} \quad \frac{x}{d}$$

$$\frac{y}{z} \quad \frac{y}{d}$$

$$\frac{x}{}\quad \mathbb{G}\frac{y}{}\quad \frac{z}{}$$

$$\frac{x}{}$$

$$\frac{z}{}$$

$$\frac{y}{}$$

$$\frac{y}{}g\quad y\quad vw\quad F$$

$$y\quad 3.d$$

$$F$$

$$y$$

$$\tau\quad \frac{x}{w}\quad x\frac{x}{}\quad y\frac{x}{}\quad z\frac{x}{}\quad \frac{s}{}\quad \frac{x}{}\quad \frac{x}{}\quad \frac{x}{}\quad \tau\quad C$$

$$\frac{x}{w}$$

$$x\frac{x}{}\quad x\frac{x}{}$$

$$z\frac{x}{}\quad \frac{x}{}$$

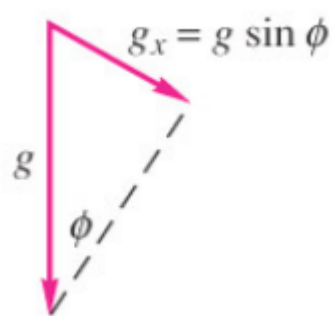
$$\tau$$

$$y\quad y\frac{x}{}$$

$$\frac{s}{}\quad \frac{x}{}\quad \tau$$

$$\frac{x}{}\quad \frac{s}{}\quad \tau$$

$$\mathfrak{vlq}$$



$$v_{\text{q}} = \frac{g}{g}$$



$$\frac{x}{s} = \frac{\tau}{v_w}$$

$$\frac{x}{s} = \frac{\tau}{F}$$

x

$$x = \frac{s}{F} \tau$$

$$x_3 = x_d$$

$$x_3 = \frac{s}{F} \tau_j$$

F

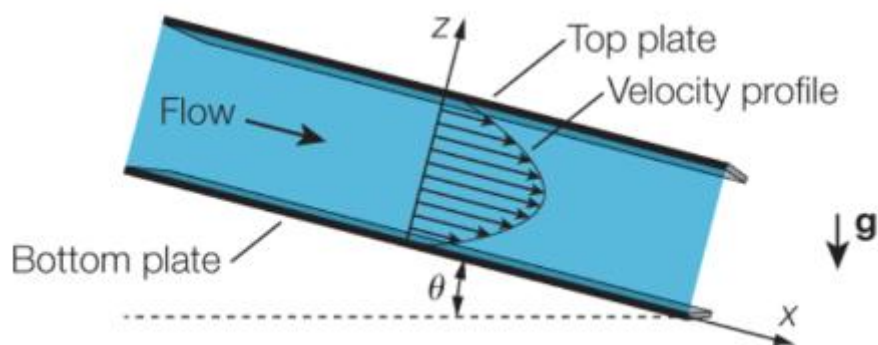
$$x \quad d \quad - \frac{s \quad \tau \quad j \quad d}{F \quad d}$$

$$F \quad - \frac{s \quad \tau \quad j \quad d}{F \quad d}$$

$F \quad F$

$$x \quad - \frac{s \quad \tau}{d}$$

$$x \quad - \frac{s \quad \tau}{d}$$



Escoamento plenamente desenvolvido dominado pelas forças friccionais

$$\frac{V}{s} = \frac{s}{d}$$